EXERCISES COURSE

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1. First class (9.19)

Problem 1.1 (1.3)**.** *For* $\forall\, n \in \mathbb{N}$*, let* $A_n = \left[-1 + \frac{1}{2n}, 1 - \frac{1}{n}\right]$*, prove* $\bigcup\limits_{n=1}^{\infty} A_n = (-1, 1)$*.*

Proof. Firstly, by definition we have that

$$\forall\, x \in \bigcup_{n=1}^{\infty} A_n, \quad \exists\, n_0 \in \mathbb{N}, \quad s.t.\ x \in A_{n_0} \subset (-1, 1).$$

Hence

$$x \in (-1, 1),$$

i.e.

$$\bigcup_{n=1}^{\infty} A_n \subset (-1, 1). \tag{1.1}$$

On the other hand, $\forall\, x \in (-1, 1)$, set

$$N = \left[\max\left\{\frac{1}{2(x+1)}, \frac{1}{1-x}\right\}\right] + 1,$$

then there is $x \in A_N$, thus $x \in \bigcup\limits_{n=1}^{\infty} A_n$, which means

$$(-1, 1) \subset \bigcup_{n=1}^{\infty} A_n. \tag{1.2}$$

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Combining (1.1) and (1.2) yields

$$\bigcup_{n=1}^{\infty} A_n = (-1, 1).$$

□

Problem 1.2 (1.5)**.** *Prove* $A = \left\{n \sin \frac{n\pi}{2}; n \in \mathbb{Z}\right\}$ *is unbounded.*

Proof. **Definition:** A set X is unbounded if and only if $\forall\, M > 0$, there exists a $x \in X$, such that $|x| > M$. Then $\forall\, M > 0$, let

$$N = 2([M] + 1) + 1,$$

hence

$$\left| N \sin \frac{N\pi}{2} \right| = 2([M] + 1) + 1 > M,$$

which implies that A is unbounded. □

Problem 1.3 (1.23)**.** *Suppose* $f(x)$ *is defined on* E*, where* $|E| \geq 3$*. Prove* $f(x)$ *is strictly monotonic on* E *iff for* $\forall\, x_1, x_2, x_3 \in E$*, if* $x_1 < x_2 < x_3$*, then there must be*

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) > 0. \tag{1.3}$$

Proof. " only if ": Obviously.

" if ": Suppose that $f(x)$ is not a strictly monotonic function, hence there exist x_1, x_2, x_3, $x_4 \in E$ with $x_1 < x_2$ and $x_3 < x_4$ such that

$$f(x_1) \leq f(x_2) \quad \text{and} \quad f(x_3) \geq f(x_4).$$

If $x_1 < x_2 < x_3 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) \quad \text{and} \quad (f(x_2) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

If $x_1 < x_3 < x_2 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_4)) \quad \text{and} \quad (f(x_1) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

Similar for other cases. Then we obtain contradictions for all cases, wihch means that $f(x)$ is strictly monotonic on E. □

Problem 1.4 (1.25)**.** *Prove* $\sin(x^2 + x)$ *is not a periodic function.*

Proof. Suppose $T > 0$ is the period of $f(x) := \sin(x^2 + x)$, we have that

$$f(T) = f(-T) = f(0) = 0.$$

Hence there are $k_1, k_2 \in \mathbb{Z}$ such that

$$T^2 + T = k_1\pi \quad \text{and} \quad T^2 - T = k_2\pi,$$

which yields

$$T = \frac{k_1 - k_2}{2}\pi.$$

Then

$$\frac{(k_1-k_2)^2}{4}\pi+\frac{k_1-k_2}{2}=k_1,$$

i.e.

$$\pi=\frac{2(k_1+k_2)}{(k_1-k_2)^2}\in\mathbb{Q},$$

contradiction. □

Problem 1.5 (1.26)**.** *Suppose $f(x)$ is defined on $(0,+\infty)$, $x_1,x_2>0$. Prove*

(1) *If $\frac{f(x)}{x}$ is decreasing, then $f(x_1+x_2)\le f(x_1)+f(x_2)$;*
(2) *If $\frac{f(x)}{x}$ is increasing, then $f(x_1+x_2)\ge f(x_1)+f(x_2)$.*

Proof. (1) Since $\frac{f(x)}{x}$ is decreasing, then $f(x_1+x_2)\le f(x_1)+f(x_2)$, we have

$$\frac{f(x_1+x_2)}{x_1+x_2}\le\frac{f(x_1)}{x_1}\quad\text{and}\quad\frac{f(x_1+x_2)}{x_1+x_2}\le\frac{f(x_2)}{x_2}.$$

Hence

$$f(x_1+x_2)=\frac{x_1}{x_1+x_2}f(x_1+x_2)+\frac{x_2}{x_1+x_2}f(x_1+x_2)\le f(x_1)+f(x_2).$$

Similar for (2). □

Problem 1.6 (1.27)**.** *Suppose $f(x)$ is defined on $(-\infty,+\infty)$, and $f(f(x))\equiv x$.*

(1) *Is $f(x)$ unique? If not, please give an example;*
(2) *If $f(x)$ is strictly increasing, is it unique? Why?*

Solution. (1) $f(x)$ is not unique. For example $f_1(x)=x$, $f_2(x)=-x$, $f_3(x)=\frac{1}{x}, x\neq 0; f_3(0)=0$.

(2) We prove that $f(x)=x$. Suppose not, if there is a $x_0\in\mathbb{R}$, such that $f(x_0)\neq x_0$. Without loss of generality, we assume that $f(x_0)>x_0$. Then $f(f(x_0))>f(x_0)$ by f is strictly increasing. Hence

$$x_0=f(f(x_0))>f(x_0)>x_0,$$

contradiction. □

Problem 1.7. *Prove that $f(x)=\sin x+\sin\sqrt{2}x$, $x\in\mathbb{R}$ is not a periodic function.*

Proof. Note that

$$f''(x)=-\sin x-2\sin\sqrt{2}x.$$

Hence we have

$$\begin{aligned}f(x)+f''(x)&=-\sin\sqrt{2}x,\\ 2f(x)+f''(x)&=\sin x.\end{aligned}$$

If $T>0$ is the period of $f(x)$, then T is also the period of $f(x)+f''(x)$ and $2f(x)+f''(x)$. Hence

$$T=\frac{2k\pi}{\sqrt{2}}=2m\pi,\quad\text{for some } k,m\in\mathbb{N},$$

which implies

$$\sqrt{2} = \frac{k}{m} \in \mathbb{Q},$$

contradiction. $\square$

Problem 1.8. *Suppose $f(x)$ is an increasing function defined on closed interval $[a, b]$, and f satisfies $f(a) \geq a$, $f(b) \leq b$. Prove that there exists $x_0 \in [a, b]$, such that $f(x_0) = x_0$.*

Proof. Set

$$A = \{x \in [a, b] : f(x) \geq x\}.$$

Obviously, A is not empty since $f(a) \geq a$, and b is an upper bound of A. Hence $\sup A$ exists. We denote $x_0 := \sup A$. For $\forall\, x \in A$, there is $x \leq x_0$. Since f is increasing, we have $x \leq f(x) \leq f(x_0)$, i.e. $f(x_0)$ is an upper bound of A, thus $x_0 \leq f(x_0)$. If $x_0 < f(x_0)$, then $f(x_0) \leq f(f(x_0))$, which means $f(x_0) \in A$, contradiction. Hence $f(x_0) = x_0$. $\square$

Problem 1.9. *Is there a function, whose period are all rational numbers but none of irrational number?*

Solution. Dirichlet function, i.e.

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \backslash \mathbb{Q}. \end{cases}$$

$\square$

2. Second class (9.26)

Problem 2.1 (2.3(5))**.** *Using definition to prove*

$$\lim_{n\to\infty} n^3 q^n = 0\,(|q| < 1).$$

Proof. It suffices to prove $\lim\limits_{n\to\infty} n^3|q|^n = 0$, and it is obviously valid when $|q| = 0$, hence we assume that $|q| \neq 0$ in the following. Let

$$\frac{1}{|q|} = 1 + \alpha.$$

By binomial theorem, we have

$$\frac{1}{|q|^n} = (1+\alpha)^n = \sum_{k=0}^{n} C_n^k \alpha^k \geq \frac{n(n-1)(n-2)(n-3)}{4!}\alpha^4$$

provided with $n \geq 4$. Hence for $n \geq 4$,

$$n^3|q|^n \leq \frac{24n^2}{(n-1)(n-2)(n-3)\alpha^4} < \frac{72}{\alpha^4}\frac{1}{n-3}.$$

Then $\forall\, \varepsilon > 0$, choosing $N = \left[\frac{72}{\alpha^4\varepsilon}\right] + 4$, for any $n > N$, there is $n^3|q|^n < \varepsilon$, i.e. $\lim\limits_{n\to\infty} n^3|q|^n = 0$. □

Problem 2.2 (2.4)**.** *Suppose for all $n \in \mathbb{N}$, there is $x_n \leq a \leq y_n$ and $\lim\limits_{n\to\infty}(y_n - x_n) = 0$. Prove*

$$\lim_{n\to\infty} x_n = a = \lim_{n\to\infty} y_n.$$

Proof. Since $x_n \leq a \leq y_n$, we know that

$$0 \leq a - x_n \leq y_n - x_n.$$

By Sandwich Theorem, we have

$$\lim_{n\to\infty} x_n = a.$$

Similar for y_n. □

Problem 2.3. *Suppose $\{a_n\}$ is monotonic increasing, $\{b_n\}$ is monotonic decreasing, and $\lim\limits_{n\to\infty}(b_n - a_n) = 0$. Prove that $\lim\limits_{n\to\infty} a_n$, $\lim\limits_{n\to\infty} b_n$ exist, and they are equal.*

Proof. Assume $\lim\limits_{n\to\infty} a_n$ doesn't exist, since $\{a_n\}$ is monotonic increasing, we know there is $\lim\limits_{n\to\infty} a_n = +\infty$. By $\{b_n\}$ is monotonic decreasing, and $\lim\limits_{n\to\infty}(b_n - a_n) = 0$, there must be $\lim\limits_{n\to\infty} b_n = -\infty$. Hence, for $\forall\, M > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N$, there are

$$a_n > M \quad \text{and} \quad b_n < -M.$$

Then

$$b_n - a_n < -2M,$$

which contradicts with $\lim_{n\to\infty}(b_n - a_n) = 0$. Thus, $\lim_{n\to\infty} a_n$ exists. It's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. $\square$

Problem 2.4 (2.6(3))**.** *$\{F_n\}$ is the Fibonacci sequence, defined by*

$$F_0 = F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}.$$

Prove $\lim_{n\to\infty} F_n = +\infty$ by definition.

Proof. **First way:** We can calculate the general terms of F_n by 'eigenvalue method' (See Problem 2.10 for details). Then we know

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right].$$

Since $\frac{1+\sqrt{5}}{2} > 1$ and $\frac{\sqrt{5}-1}{2} < 1$, we have $\lim_{n\to\infty} F_n = +\infty$.

Second way: We can prove by induction that $F_n \geq n$ for $n \geq 1$. Since $\lim_{n\to\infty} n = +\infty$, we know $\lim_{n\to\infty} F_n = +\infty$. $\square$

Problem 2.5 (2.10(6))**.** *Calculate*

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}).$$

Solution.

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}) = \lim_{n\to\infty} \frac{\sqrt[3]{n}}{(\sqrt[3]{n+1})^2 + \sqrt[3]{n+1}\sqrt[3]{n} + (\sqrt[3]{n})^2} = 0.$$

$\square$

Problem 2.6 (2.14)**.** *Calculate $\lim_{n\to\infty} x_n$.*

(1) $x_n = \dfrac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$;

(2) $x_n = \displaystyle\sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}}$;

(3) $x_n = \sqrt[n]{n\ln n}$.

Solution. (1)Note that

$$\frac{q}{p} < \frac{q+1}{p+1}, \quad \text{for } 0 < q < p.$$

Then

$$x_n = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$$

$$
\begin{aligned}
&= \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n} \\
&< \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdots\frac{2n-2}{2n-1}\cdot\frac{2n}{2n+1} \\
&= \frac{2}{1}\cdot\frac{4}{3}\cdot\frac{6}{5}\cdots\frac{2n}{2n-1}\cdot\frac{1}{2n+1} \\
&= \frac{1}{x_n}\frac{1}{2n+1},
\end{aligned}
$$

which gives us

$$x_n < \frac{1}{\sqrt{2n+1}}.$$

Hence $\lim_{n\to\infty} x_n = 0$.

(2) Note that $(n+1)^2 - n^2 + 1 = 2(n+1)$, we have

$$2 = \frac{2(n+1)}{n+1} \le \sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}} \le \frac{2(n+1)}{n} \to 2, \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 2$.

(3) When $n > 3$, there is $n < n\ln n < n^2$. Then

$$1 \leftarrow \sqrt[n]{n} < \sqrt[n]{n\ln n} < \sqrt[n]{n^2} \to 1, \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 1$. □

Problem 2.7 (2.17)**.** *Sequence $\{q_n\}$ satisfies*

$$0 < q_n < 1, \quad (1-q_n)q_{n+1} > \frac{1}{4}, \quad \forall\, n \in \mathbb{N}. \tag{2.1}$$

Prove that $\{q_n\}$ is monotonic increasing and $\lim_{n\to\infty} q_n = \frac{1}{2}$.

Proof. By (2.1) and the mean value inequality, we have

$$\frac{q_{n+1}}{q_n} > \frac{1}{4q_n(1-q_n)} \ge \frac{1}{4\left(\frac{q_n+1-q_n}{2}\right)^2} = 1,$$

i.e. $q_{n+1} > q_n$. Hence $\{q_n\}$ is monotonic increasing. Then by the monotone bounded convergence theorem, we know $\lim_{n\to\infty} q_n$ exists. Denote $q := \lim_{n\to\infty} q_n$, by (2.1),

$$(1-q)q \ge \frac{1}{4},$$

i.e.

$$\left(q - \frac{1}{2}\right)^2 \le 0,$$

which means $q = \frac{1}{2}$. □

Problem 2.8 (2.19)**.** *Suppose that* $0 < a_1 < b_1$*, let*

$$a_{n+1} = \sqrt{a_n b_n}, \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist, and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$*.*

Proof. By the mean value inequality, we have

$$a_{n+1} = \sqrt{a_n b_n} \le \frac{1}{2}(a_n + b_n) = b_{n+1} \quad \text{for } n \ge 1.$$

Hence

$$a_{n+1} = \sqrt{a_n b_n} \ge a_n \quad \text{and} \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \le b_n \quad \text{for } n \ge 2.$$

Then

$$a_2 \le \cdots \le a_n \le b_n \le \cdots \le b_2 \quad \text{for all } n \ge 2.$$

By the monotone bounded convergence theorem, we know $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, and it's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. □

Problem 2.9. *Suppose that* $a_1 > b_1 > 0$*, let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{2a_n b_n}{a_n + b_n} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist, and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$*.*

Proof. The existence is similar to Problem 2.8. Thus, we only need to prove $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$, and if we notice that

$$a_{n+1} b_{n+1} = a_n b_n = \cdots = a_1 b_1,$$

and

$$a_{n+1} \ge \sqrt{a_n b_n} \ge b_{n+1},$$

it's easy to obtain the conclusion. □

Problem 2.10. *Suppose that* $a_1 = \alpha$*,* $b_1 = \beta$*. Let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{a_{n+1} + b_n}{2} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist,* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$*, and find the limitation.*

Proof. By $a_{n+1} = (a_n + b_n)/2$, we have $b_n = 2a_{n+1} - a_n$. Hence

$$2a_{n+2} - a_{n+1} = \frac{a_{n+1} + 2a_{n+1} - a_n}{2} = \frac{3}{2}a_{n+1} - \frac{1}{2}a_n,$$

which is

$$a_{n+2} = \frac{5}{4}a_{n+1} - \frac{1}{4}a_n. \tag{2.2}$$

Then the characteristic equation of (2.2) is

$$x^2 - \frac{5}{4}x + \frac{1}{4} = 0.$$

We find that $x = 1$ and $x = \frac{1}{4}$ are the solution. Hence the general form of a_n is

$$a_n = A + B\left(\frac{1}{4}\right)^n.$$

By $a_1 = \alpha$, $a_2 = (a_1 + b_1)/2 = (\alpha + \beta)/2$, we have

$$a_n = \frac{1}{3}\alpha + \frac{2}{3}\beta + \frac{2}{3}(\alpha - \beta)\left(\frac{1}{4}\right)^{n-1}.$$

Hence $\lim\limits_{n\to\infty} a_n$ and $\lim\limits_{n\to\infty} b_n$ exist, and $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = \frac{1}{3}\alpha + \frac{2}{3}\beta$. □

Problem 2.11 (2.20)**.** *Calculate following limitations.*

(2) $\lim\limits_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n$;

(3) $\lim\limits_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2}$.

Solution. (2)We first note that

$$\left(1 + \frac{1}{n^2}\right)^{n^2} < e.$$

(This can be proved by definition of e.) Then we have

$$1 < \left(1 + \frac{1}{n^2}\right)^n < e^{\frac{1}{n}} \to 1 \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem

$$\lim_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n = 1.$$

(3)**First way:** By binomial theorem, we know

$$\left(1 + \frac{1}{n}\right)^{n^2} = 1 + n^2\frac{1}{n} + \cdots \geq n.$$

Hence

$$\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2} = +\infty.$$

Second way: By the definition of e, we know there exists a $N \in \mathbb{N}$, such that for $n > N$, there is

$$\frac{e}{2} < \left(1 + \frac{1}{n}\right)^n < e.$$

(Via choosing $\varepsilon = \frac{e}{2}$.) Hence we have

$$\left(1+\frac{1}{n}\right)^{n^2} > \left(\frac{e}{2}\right)^n \to +\infty \quad \text{as } n \to \infty,$$

i.e.

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} = +\infty.$$

□

Problem 2.12. *Using* $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$ *to prove* $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e$.

Proof. By binomial theorem, we have

$$\begin{aligned}
\left(1+\frac{1}{n}\right)^n &= \sum_{k=0}^{n} C_n^k \frac{1}{n^k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{n(n-1)\cdots(n-k+1)}{k!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) \\
&\le \sum_{k=0}^{n} \frac{1}{k!}.
\end{aligned}$$

From the above calculation, we know for $\forall\, m > n$, there is

$$\left(1+\frac{1}{m}\right)^m \ge \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{m}\right)\cdots\left(1-\frac{k-1}{m}\right).$$

Let $m \to +\infty$, there is

$$e \ge \sum_{k=0}^{n} \frac{1}{k!} \ge \left(1+\frac{1}{n}\right)^n.$$

Hence

$$\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e.$$

□

Problem 2.13. *Calculate the limitation of*

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right).$$

Solution.

$$\lim_{n\to\infty}\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)=\lim_{n\to\infty}\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdots\frac{n-1}{n}\cdot\frac{n+1}{n}=\frac{1}{2}.$$

□

Problem 2.14. *Suppose* $\lim_{n\to\infty} a_n = a$. *Prove*

$$\lim_{n\to\infty}\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}=a,$$

where $p_k>0$ *and* $\lim_{n\to\infty}\dfrac{p_n}{p_1+p_2+\cdots+p_n}=0$.

Proof. Without loss of generality, we can assume $a=0$. (Otherwise, we can consider a_n-a instead.) Since $\lim_{n\to\infty} a_n=0$, we know for $\forall\,\varepsilon>0$, there exists $N_1\in\mathbb{N}$, such that for all $n>N_1$, there is $|a_n|<\varepsilon/2$. Then

$$\left|\frac{p_1a_n+\cdots+p_{n-N_1}a_{N_1+1}}{p_1+p_2+\cdots+p_n}\right|\le\frac{p_1+\cdots+p_{n-N_1}}{p_1+p_2+\cdots+p_n}\cdot\frac{\varepsilon}{2}<\frac{\varepsilon}{2}.$$

Since a_n is convergence, we know $\{a_n\}$ is bounded. We assume $|a_n|\le M$ for some $M>0$. By $\lim_{n\to\infty}\dfrac{p_n}{p_1+p_2+\cdots+p_n}=0$, we have

$$\begin{aligned}\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|&\le\left(\frac{p_{n-N_1+1}+\cdots+p_n}{p_1+p_2+\cdots+p_n}\right)M\\&<M\sum_{k=1}^{N_1}\frac{p_{n-N_1+k}}{p_1+p_2+\cdots+p_{n-N_1+k}}\\&\to 0\quad\text{as } n\to\infty.\end{aligned}$$

Hence there exists a $N_2\in\mathbb{N}$ such that for all $n>N_2$, there is

$$\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|<\frac{\varepsilon}{2}.$$

Then, set $N=\max\{N_1,N_2\}$, we know for all $n>N$, there is

$$\begin{aligned}\left|\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|&\le\left|\frac{p_1a_n+\cdots+p_{n-N_1}a_{N_1+1}}{p_1+p_2+\cdots+p_n}\right|+\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|\\&<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon,\end{aligned}$$

i.e.

$$\lim_{n\to\infty}\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}=0.$$

□

Problem 2.15. *Define sequence* $\{a_n\}$ *by* $a_{n+1}=2a_n-a_n^2$, *where* a_0 *is given. Discuss the convergence and divergence of* $\{a_n\}$ *regards the choice of* a_0.

Solution. Firstly, we have

$$1 - a_{n+1} = 1 - 2a_n + a_n^2 = (1 - a_n)^2.$$

By induction, we have

$$1 - a_{n+1} = (1 - a_n)^2 = \cdots = (1 - a_0)^{2^{n+1}},$$

i.e.

$$a_n = 1 - (1 - a_0)^{2^n} \quad \text{for } n \geq 1.$$

Hence,

When $|1 - a_0| < 1$, i.e. $0 < a_0 < 2$, $\{a_n\}$ is convergent, and $\lim\limits_{n\to\infty} a_n = 1$.

When $|1 - a_0| = 1$, i.e. $a_0 = 0$ or $a_0 = 2$, $\{a_n\}$ is convergent, and $\lim\limits_{n\to\infty} a_n = 0$.

When $|1 - a_0| > 1$, i.e. $a_0 < 0$ or $a_0 > 2$, $\{a_n\}$ is divergent, $\lim\limits_{n\to\infty} a_n = -\infty$. $\square$

3. Third class (10.10)

Problem 3.1 (2.21)**.** *Suppose that* $\{b_n\}$ *is strictly increasing and* $\lim_{n\to\infty} b_n = +\infty$*. Prove that if*

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A, \tag{3.1}$$

where A *is finite or* $\pm\infty$*. Then*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We first prove the case for A is finite. By (3.1), we know $\forall\, \varepsilon > 0$, there exists a $N_1 \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\left| \frac{a_n - a_{n-1}}{b_n - b_{n-1}} - A \right| < \varepsilon.$$

Since $b_n > b_{n-1}$ for all $n \in \mathbb{N}$, we have

$$(A - \varepsilon)(b_n - b_{n-1}) < a_n - a_{n-1} < (A + \varepsilon)(b_n - b_{n-1}).$$

For given N_1, summing all those inequalities, we obtain

$$(A - \varepsilon)(b_n - b_{N_1}) < a_n - a_{N_1} < (A + \varepsilon)(b_n - b_{N_1}),$$

i.e.

$$\left| \frac{a_n - a_{N_1}}{b_n - b_{N_1}} - A \right| < \varepsilon.$$

Note the identity

$$\frac{a_n}{b_n} - A = \left(1 - \frac{b_{N_1}}{b_n}\right) \cdot \left(\frac{a_n - a_{N_1}}{b_n - b_{N_1}} - A\right) + \frac{a_{N_1} - Ab_{N_1}}{b_n},$$

and combining $\lim_{n\to\infty} b_n = +\infty$, we know there exists a $N_2 \in \mathbb{N}$, such that for $n > N_2$, there is

$$0 < 1 - \frac{b_{N_1}}{b_n} < 2 \quad \text{and} \quad \left| \frac{a_{N_1} - Ab_{N_1}}{b_n} \right| < \varepsilon.$$

Choosing $N = \max\{N_1, N_2\}$, then for $\forall\, n > N$, we have

$$\left| \frac{a_n}{b_n} - A \right| < 3\varepsilon,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Next, we prove the case for $A = +\infty$. By (3.1), we know $\forall\, M > 0$, there exists a $N_1 \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\frac{a_n - a_{n-1}}{b_n - b_{n-1}} > 3M.$$

Since $b_n > b_{n-1}$ for all $n \in \mathbb{N}$, we have

$$a_n - a_{n-1} > 3M(b_n - b_{n-1}).$$

For given N_1, summing all those inequalities, we obtain

$$a_n - a_{N_1} > 3M(b_n - b_{N_1}),$$

i.e.

$$\frac{a_n - a_{N_1}}{b_n - b_{N_1}} > 3M.$$

Note the identity

$$\frac{a_n}{b_n} = \left(1 - \frac{b_{N_1}}{b_n}\right) \cdot \left(\frac{a_n - a_{N_1}}{b_n - b_{N_1}}\right) + \frac{a_{N_1}}{b_n},$$

and combining $\lim_{n\to\infty} b_n = +\infty$, we know there exists a $N_2 \in \mathbb{N}$, such that for $n > N_2$, there is

$$\frac{1}{2} < 1 - \frac{b_{N_1}}{b_n} \quad \text{and} \quad \left|\frac{a_{N_1}}{b_n}\right| < \frac{1}{2}M.$$

Choosing $N = \max\{N_1, N_2\}$, then for $\forall\, n > N$, we have

$$\frac{a_n}{b_n} > \frac{3}{2}M - \frac{1}{2}M = M,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = +\infty.$$

Similar for $A = -\infty$. □

Problem 3.2 ($\frac{0}{0}$ type of Stolz theorem)**.** *Suppose that* $\lim_{n\to\infty} a_n = 0$, $\lim_{n\to\infty} b_n = 0$, *and* $\{b_n\}$ *is strictly decreasing. Prove that if*

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A, \tag{3.2}$$

where A is finite or $\pm\infty$. Then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We only prove the case for A is finite. By (3.2), we know $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\left|\frac{a_n - a_{n-1}}{b_n - b_{n-1}} - A\right| < \varepsilon.$$

Since $b_n > b_{n+1}$ for all $n \in \mathbb{N}$, we have

$$(A - \varepsilon)(b_n - b_{n+1}) < a_n - a_{n+1} < (A + \varepsilon)(b_n - b_{n+1}).$$

For any $m > n$, summing all those inequalities, we obtain

$$(A - \varepsilon)(b_n - b_m) < a_n - a_m < (A + \varepsilon)(b_n - b_m),$$

i.e.

$$\left|\frac{a_n - a_m}{b_n - b_m} - A\right| < \varepsilon.$$

Let $m \to \infty$, and combining $\lim\limits_{n\to\infty} a_n = 0$, $\lim\limits_{n\to\infty} b_n = 0$, we know for $\forall\, n > N$, there is

$$\left|\frac{a_n}{b_n} - A\right| \le \varepsilon,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

□

Remark 3.3. *If $A = \infty$ in Problem 3.1 and Problem 3.2, the conclusion is not correct. For example $a_n = (-1)^n n$, $b_n = n$ in Problem 3.1 and $a_n = (-1)^n \frac{1}{n}$, $b_n = \frac{1}{n}$ Problem 3.2.*

Problem 3.4. *Using Stolz theorem to prove*

(1) *If $\lim\limits_{n\to\infty} a_n = a$, then $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k$ exists and $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k = a$.*

(2) *If $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k = a$, and $\lim\limits_{n\to\infty} n(a_n - a_{n-1}) = 0$, then $\lim\limits_{n\to\infty} a_n$ exists and $\lim\limits_{n\to\infty} a_n = a$.*

Proof. (1) By Stolz theorem,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} a_k = \lim_{n\to\infty} \frac{\sum\limits_{k=1}^{n} a_k - \sum\limits_{k=1}^{n-1} a_k}{n-(n-1)} = \lim_{n\to\infty} a_n = a.$$

(2) Assume $a_0 = 0$. Let $A_n := a_n - a_{n-1}$, then $a_n = \sum\limits_{k=1}^{n} A_k$. The conditions become

$$a = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} a_k = \lim_{n\to\infty} \frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n},$$

and

$$0 = \lim_{n\to\infty} n(a_n - a_{n-1}) = \lim_{n\to\infty} nA_n.$$

Hence

$$\begin{aligned}
\lim_{n\to\infty} a_n &= \lim_{n\to\infty} \frac{na_n}{n} = \lim_{n\to\infty} \frac{n\sum\limits_{k=1}^{n} A_k}{n} \\
&= \lim_{n\to\infty} \frac{nA_1 + nA_2 + \cdots + nA_n}{n} \\
&= \lim_{n\to\infty} \left(\frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n} + \frac{A_2 + \cdots + (n-1)A_n}{n}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \lim_{n\to\infty} \frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n} + \lim_{n\to\infty} \frac{A_2 + \cdots + (n-1)A_n}{n} \\
&= a + \lim_{n\to\infty} \frac{A_2 + \cdots + (n-1)A_n}{n} \\
&= a + \lim_{n\to\infty} \frac{(n-1)A_n}{n-(n-1)} \qquad \text{(Stolz theorem)} \\
&= a + \lim_{n\to\infty} \frac{n-1}{n} \cdot \lim_{n\to\infty} nA_n \\
&= a.
\end{aligned}
$$

□

Problem 3.5. *Suppose that* $x_{n+1} = x_n(1-x_n)$, $n = 1, 2, \cdots$, $0 < x_1 < 1$. *Prove that* $\lim_{n\to\infty} x_n = 0$ *and* $\lim_{n\to\infty} nx_n = 1$.

Proof. Firstly. note that

$$x_{n+1} - x_n = -x_n^2 \leq 0.$$

Then $\{x_n\}$ is monotonic decreasing. Since $0 < x_1 < 1$, we can prove $0 < x_n < 1$ by induction. Hence by monotone bounded convergence theorem, we know $\lim_{n\to\infty} x_n$ exists and $\lim_{n\to\infty} x_n = 0$. And by Stolz theorem, we have

$$
\begin{aligned}
\lim_{n\to\infty} nx_n &= \lim_{n\to\infty} \frac{n}{\frac{1}{x_n}} = \lim_{n\to\infty} \frac{n-(n-1)}{\frac{1}{x_n} - \frac{1}{x_{n-1}}} \\
&= \lim_{n\to\infty} \frac{x_{n+1}x_n}{x_n - x_{n+1}} = \lim_{n\to\infty} \frac{x_{n+1}}{x_n} \\
&= \lim_{n\to\infty} (1 - x_n) = 1.
\end{aligned}
$$

□

Remark 3.6. *If we replace* $x_{n+1} = x_n(1-x_n)$ *by* $x_{n+1} = \ln(1+x_n)$, *we can have a similar conclusion.*

Problem 3.7 (2.23)**.** *Suppose* f *is defined on* (a,b), *and for* $\forall\, \xi \in (a,b)$, *there exists a* $\delta > 0$, *such that for* $x \in (\xi - \delta, \xi + \delta) \cap (a,b)$,

(1) *If* $x < \xi$, *there is* $f(x) < f(\xi)$;
(2) *If* $x > \xi$, *there is* $f(x) > f(\xi)$.

Prove that f *is strictly increasing in* (a,b).

Proof. For $\forall\, x_1, x_2 \in (a,b)$ with $x_1 < x_2$, we want to show $f(x_1) < f(x_2)$. By assumption, $\forall\, \xi \in [x_1, x_2]$, there exists a $\delta = \delta(\xi)$ (If necessary, we can shrink δ so that $U(\xi,\delta) \subset (a,b)$), such that for $x \in U(\xi,\delta)$,

(1) If $x < \xi$, there is $f(x) < f(\xi)$;
(2) If $x > \xi$, there is $f(x) > f(\xi)$.

Then we know

$$[x_1, x_2] \subset \bigcup_{\xi \in [x_1, x_2]} U(\xi, \delta(\xi))$$

is an open covering. By Heine–Borel theorem, there exists a finite subcovering, i.e. there are $\xi_1, \cdots, \xi_n \in [x_1, x_2]$ and $\delta_1, \cdots, \delta_n > 0$, such that

$$[x_1, x_2] \subset \bigcup_{i=1}^{n} U(\xi_i, \delta_i).$$

Without loss of generality, we assume that

$$\xi_1 < \xi_2 < \cdots < \xi_n \quad \text{and} \quad U(\xi_i, \delta_i) \cap U(\xi_{i+1}, \delta_{i+1}) \neq \emptyset, \ i = 1, 2, \cdots, n-1.$$

Hence, we have $f(x_1) \leq f(\xi_1) < f(\xi_2) < \cdots < f(\xi_n) \leq f(x_2)$, i.e. $f(x_1) < f(x_2)$ valids for any $x_1 < x_2$. Thus, f is strictly increasing in (a, b). □

Problem 3.8 (2.25)**.** *Using supremum and infimum principle to prove accumulation point principle.*

Proof. **First way:** Suppose S is a bounded set consists of infinite numbers of elements. By supremum and infimum principle, we know that $\sup S$ and $\inf S$ exist. If one of them is not a isolated point of S, it's obviously a accumulation point. Now, we assume none of them is the accumulation point of S. Set

$$E := \{x \in \mathbb{R} \,|\text{There are only a finite number of elments in } S \text{ that are less than } x\}.$$

Then E is nonempty and has an upper bound. Let $\eta := \sup E$, we prove η is a accumulation point of S. Indeed, by the construction of E, we know that $\forall\, \varepsilon > 0$, there must be $\eta + \varepsilon \notin E$, i.e. there are infinite numbers of elements in S that are less that $\eta + \varepsilon$. Since there eixsts a $x_0 \in E$ such that $\eta - \varepsilon < x_0$, we know that there are only a finite number of elments in S that are less than $\eta - \varepsilon$. Then $(\eta - \varepsilon, \eta + \varepsilon)$ contains infinite numbers of elements in S, which means that η is a accumulation point of S.

Second way: We first claim that any sequence in $\mathbb{R}$ has at least a monotonic subsequence (either increasing or decreasing). Indeed, if there is no increasing subsequence in $\{x_n\}$, we know there exists a $n_1 > 0$, such that $\forall\, n > n_1$, there is $x_n < x_{n_1}$. Similarly, there is no increasing subsequence in $\{x_n\}_{n > n_1}$, we know there exists a $n_2 > n_1$, such that $\forall\, n > n_2$, there is $x_n < x_{n_2} < x_{n_1}$. Proceeding like this, we can find a strictly decreasing subsequence $\{x_{n_k}\}$. Then it's easy to see that accumulation point principle is valid. □

Problem 3.9 (2.34)**.** *Prove that if $x_n > 0$ and*

$$\overline{\lim_{n\to\infty}}\, x_n \cdot \overline{\lim_{n\to\infty}}\, \frac{1}{x_n} = 1,$$

then the sequence $\{x_n\}$ is convergent.

Proof. Suppose there is a subsequence $\{x_{n_k}\} \subset \{x_n\}$, such that

$$\lim_{k\to\infty} x_{n_k} = \underline{\lim_{n\to\infty}}\, x_n.$$

By the definition of limit superior and $x_n > 0$, we know

$$\frac{1}{x_{n_k}} \leq \sup_{n \geq n_k} \frac{1}{x_n}.$$

Hence

$$\varliminf_{n\to\infty} x_n = \lim_{k\to\infty} x_{n_k} \geq \lim_{k\to\infty} \frac{1}{\sup\limits_{n\geq n_k} \frac{1}{x_n}} = \frac{1}{\varlimsup\limits_{n\to\infty} \frac{1}{x_n}} = \varlimsup_{n\to\infty} x_n,$$

i.e.

$$\varliminf_{n\to\infty} x_n = \varlimsup_{n\to\infty} x_n.$$

Hence the sequence $\{x_n\}$ is convergent. □

Problem 3.10 (2.35)**.** *Suppose that $\{x_n\}$ is bounded, and*

$$\lim_{n\to\infty} (x_{n+1} - x_n) = 0. \tag{3.3}$$

Denote $l = \varliminf\limits_{n\to\infty} x_n$ and $L = \varlimsup\limits_{n\to\infty} x_n$. Prove that any number in $[l, L]$ is the limitation of a subsequence of $\{x_n\}$.

Proof. **First way:** By definition, l, L are both accumulation points of $\{x_n\}$. Then we assume $l < L$ and $a \in (l, L)$, and we will prove a is a accumulation point of $\{x_n\}$ in the following. We first prove the claim that for any given $\varepsilon > 0$ and $N \in \mathbb{N}$, there must exist a $\bar{n} > N$ such that

$$|x_{\bar{n}} - a| < \varepsilon.$$

By (3.3), we know there exists a $N' \in \mathbb{N}$, such that for $\forall\, n > N'$, there is

$$|x_{n+1} - x_n| < \varepsilon.$$

Let $N_0 := \max\{N, N'\}$. We know there must exist at least two points $x_{n'}$, $x_{n''}$ in $\{x_n\}_{n\geq N_0}$ such that $x_{n'} < a$, $x_{n''} > a$ (otherwise, if there is no points that are less than a, we must have $\varliminf\limits_{n\to\infty} x_n \geq a$, contradicts with $l < a$; if there is no points that are large than a, we must have $\varlimsup\limits_{n\to\infty} x_n \leq a$, contradicts with $a < L$). Without loss of generality, we assume $n' < n''$. Let $\bar{n}$ be the maximal integer which satisfies $n' \leq n \leq n''$ and $x_n \leq a$. Clearly, $\bar{n} \leq n'' - 1$ and $x_{\bar{n}} \leq a$, $x_{\bar{n}+1} > a$. Hence $\bar{n} > N$, and

$$|x_{\bar{n}} - a| \leq x_{\bar{n}+1} - x_{\bar{n}} < \varepsilon.$$

The claim is proved.

Now choosing $\varepsilon_1 = 1$, $N_1 = 1$, there exists a x_{n_1} $(n_1 > 1)$ such that $|x_{n_1} - a| < 1$. Then choosing $\varepsilon_2 = \frac{1}{2}$, $N_2 = n_1$, there exists a x_{n_2} $(n_2 > n_1)$ such that $|x_{n_2} - a| < \frac{1}{2}$. Next, choosing $\varepsilon_3 = \frac{1}{3}$, $N_3 = n_2$, there exists a x_{n_3} $(n_3 > n_2)$ such that $|x_{n_3} - a| < \frac{1}{3}$. Proceeding like this, we obtain a subsequence $\{x_{n_k}\} \subset \{x_n\}$ satisfies

$$|x_{n_k} - a| < \frac{1}{k}.$$

Hence $\lim\limits_{k\to\infty} x_{n_k} = a$, i.e. a is a accumulation point of $\{x_n\}$.

Second way: By definition, l, L are both accumulation points of $\{x_n\}$. Then we assume $l < L$ and $a \in (l, L)$, and we will prove a is a accumulation point of $\{x_n\}$ in the following. Let $\delta := \min\{L - a, a - l\}$. For $\forall\, i \in \mathbb{N}$, there exists a $k_i \in \mathbb{N}$, when $n \geq k_i$, we have $|x_{n+1} - x_n| < \frac{\delta}{2^i}$.

By definition of l, L, we can choose $l_1 \geq k_1$, such that $x_{l_1} > L - \frac{\delta}{2}$, and choose $m_1 > l_1$, such that $x_{m_1} < l + \frac{\delta}{2}$. Again, choosing $l_2 > \max\{k_2, m_1\}$, such that $x_{l_2} > L - \frac{\delta}{2^2}$, and choose $m_2 > l_2$, such that $x_{m_2} < l + \frac{\delta}{2^2}$. Proceeding like this, we can choose $l_i > \max\{k_i, m_{i-1}\}$, such that $x_{l_i} > L - \frac{\delta}{2^i}$, and choose $m_i > l_i$, such that $x_{m_1} < l + \frac{\delta}{2^i}$. Hence, we know there exists n_i, $l_i < n_i < m_i$ $(i \in \mathbb{N})$, such that

$$x_{n_i} < l + \delta \leq a, \quad x_{n_i - 1} \geq a \quad (i \in \mathbb{N}).$$

Then

$$|x_{n_i} - a| < |x_{n_i} - x_{n_i - 1}| \quad (i \in \mathbb{N}).$$

By $n_i > l_i > k_i$, we know $n_i - 1 \geq k_i$. Hence

$$|x_{n_i} - a| < |x_{n_i} - x_{n_i - 1}| < \frac{\delta}{2^i}.$$

Hence $\lim\limits_{i \to \infty} x_{n_i} = a$, i.e. a is a accumulation point of $\{x_n\}$. □

Problem 3.11 (2.36)**.** *Suppose $\{x_n\}$ and $\{y_n\}$ satisfy*

$$x_{n+1} = y_n + q x_n \, (0 < q < 1), \quad n = 1, 2, \cdots.$$

Prove $\{y_n\}$ converges iff $\{x_n\}$ converges.

Proof. "if": Note that $y_n = x_{n+1} - q x_n$, it's easy to show $\{y_n\}$ converges if $\{x_n\}$ is convergent.

"only if": We first prove that $\{x_n\}$ is bounded. Since $\{y_n\}$ is convergent, we know it is bounded. Assume that $|y_n| \leq M$ for some $M > 0$. Then

$$\begin{aligned}
|x_{n+1}| &= |y_n + q x_n| = |y_n + q(y_{n-1} + q x_{n-1})| \\
&= \left|y_n + q y_{n-1} + q^2 x_{n-1}\right| \\
&= \left|y_n + q y_{n-1} + \cdots + q^{n-1} y_1 + q^n x_1\right| \\
&\leq M\left(1 + q + \cdots + q^{n-1}\right) + q^n |x_1| \\
&< \frac{M}{1-q} + |x_1|.
\end{aligned}$$

Hence the upper limit and lower limit of $\{x_n\}$ exist. We have from $x_{n+1} = y_n + q x_n$ that

$$\overline{\lim_{n\to\infty}}\, x_n = \overline{\lim_{n\to\infty}}\,(y_{n-1} + q x_{n-1}) = \lim_{n\to\infty} y_n + q \overline{\lim_{n\to\infty}}\, x_{n-1} \leq \lim_{n\to\infty} y_n + q \overline{\lim_{n\to\infty}}\, x_n,$$

which yields

$$\overline{\lim_{n\to\infty}}\, x_n \leq (1-q)^{-1} \lim_{n\to\infty} y_n.$$

Similarly, we have

$$\underline{\lim_{n\to\infty}}\, x_n \geq (1-q)^{-1} \lim_{n\to\infty} y_n.$$

Hence

$$\overline{\lim_{n\to\infty}}\, x_n = \underline{\lim_{n\to\infty}}\, x_n = (1-q)^{-1} \lim_{n\to\infty} y_n,$$

i.e. $\{x_n\}$ converges. $\square$

Problem 3.12.

(1) *(2.37). Suppose $\{x_n\}$ satisfies for $\forall\, n, m \in \mathbb{N}$, there is*

$$0 \le x_{n+m} \le x_n + x_m.$$

Prove that $\left\{\dfrac{x_n}{n}\right\}$ has a limitation.

(2) *Suppose $\{x_n\}$ satisfies for $\forall\, n, m \in \mathbb{N}$, there is*

$$0 \le x_{n+m} \le x_n \cdot x_m.$$

Prove that $\left\{x_n^{\frac{1}{n}}\right\}$ has a limitation.

Proof. (1) By

$$x_n \le x_{n-1} + x_1 \le x_{n-2} + 2x_1 \le \cdots \le nx_1,$$

we have

$$0 \le \frac{x_n}{n} \le x_1.$$

Hence $\left\{\dfrac{x_n}{n}\right\}$ is bounded. Denote $\underline{\lim}_{n\to\infty} \dfrac{x_n}{n} = l$, then $0 \le l \le x_1$. By definition, for $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that

$$\frac{x_N}{N} < l + \varepsilon.$$

For any $n > N$, we choose $q \in \mathbb{N}$ and $0 \le r < N$, such that $n = qN + r$. Then

$$x_n = x_{qN+r} \le x_{qN} + x_r \le qx_N + rx_1 \le qx_N + Nx_1.$$

Hence

$$\frac{x_n}{n} \le \frac{qx_N}{n} + \frac{Nx_1}{n} \le \frac{x_N}{N} + \frac{Nx_1}{n} < l + \varepsilon + \frac{Nx_1}{n}.$$

Thus

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} \le l + \varepsilon.$$

Let $\varepsilon \to 0$, we have

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} \le l = \underline{\lim_{n\to\infty}}\, \frac{x_n}{n}.$$

Hence

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} = \underline{\lim_{n\to\infty}}\, \frac{x_n}{n},$$

i.e. $\lim\limits_{n\to\infty} \dfrac{x_n}{n}$ exists.

(2) The proof is similar to (1), we leave it to readers. $\square$

Problem 3.13.

(1) *Suppose* $\{x_n\}$ *is a positive sequence. Prove that*

$$\overline{\lim_{n\to\infty}}\, n\left(\frac{1+x_{n+1}}{x_n}-1\right)\geq 1.$$

(2) *Suppose* $\{x_n\}$ *is a positive sequence. Prove that*

$$\overline{\lim_{n\to\infty}}\left(\frac{x_1+x_{n+1}}{x_n}\right)^n\geq e.$$

Proof. (1) Proof by contradiction. Assume that

$$\overline{\lim_{n\to\infty}}\, n\left(\frac{1+x_{n+1}}{x_n}-1\right)<1.$$

Hence there exists a $N\in\mathbb{N}$, such that $\forall\, n\geq N$, there is

$$n\left(\frac{1+x_{n+1}}{x_n}-1\right)<1.$$

Then we have

$$\frac{1}{n+1}<\frac{x_n}{n}-\frac{x_{n+1}}{n+1},\quad n\geq N.$$

Summing all inequalities from N to n yields

$$\frac{1}{N+1}+\frac{1}{N+2}+\cdots+\frac{1}{n+1}<\frac{x_N}{N}-\frac{x_{n+1}}{n+1}\leq\frac{x_N}{N}.$$

However, we already konw $\lim\limits_{n\to\infty}\left(\frac{1}{N+1}+\frac{1}{N+2}+\cdots+\frac{1}{n+1}\right)=+\infty$, which makes a contradiction.

(2) Proof by contradiction. Assume that

$$\overline{\lim_{n\to\infty}}\left(\frac{x_1+x_{n+1}}{x_n}\right)^n<e.$$

Hence there exists a $N\in\mathbb{N}$, such that $\forall\, n\geq N$, there is

$$\left(\frac{x_1+x_{n+1}}{x_n}\right)^n<e<\left(1+\frac{1}{n-1}\right)^n.$$

Then we have

$$\frac{1}{n}<\frac{x_n}{n-1}-\frac{x_{n+1}}{n},\quad n\geq N.$$

Summing all inequalities from N to n yields

$$\frac{1}{N}+\frac{1}{N+1}+\cdots+\frac{1}{n}<\frac{x_N}{N-1}-\frac{x_{n+1}}{n}\leq\frac{x_N}{N-1}.$$

However, we already konw $\lim\limits_{n\to\infty}\left(\frac{1}{N}+\frac{1}{N+1}+\cdots+\frac{1}{n}\right)=+\infty$, which makes a contradiction. □

Remark 3.14. *Both constants in Problem 3.13 are optimal. Indeed, we can choose* $x_n=n\ln n$.

Problem 3.15. *Suppose that* $x_n > 0$. *Prove* $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} \le 1$ *iff* $\lim\limits_{n\to\infty} \dfrac{x_n}{l^n} = 0,\ \forall\, l > 1$.

Proof. "if": If $\lim\limits_{n\to\infty} \dfrac{x_n}{l^n} = 0$, we know there exists a $N \in \mathbb{N}$ such that $\forall\, n > N$, there is $x_n < \frac{1}{2} l^n$. Hence $\sqrt[n]{x_n} < \left(\frac{1}{2}\right)^{\frac{1}{n}} l$, which means $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} \le l$. Since $l > 1$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} \le 1$.

"only if": For $\forall\, l > 1$, by $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} \le 1$ we have there exists a $N \in \mathbb{N}$, such that $\forall\, n > N$, there is

$$\sqrt[n]{x_n} < 1 + \frac{l-1}{2} = \frac{l+1}{2}.$$

Then

$$\frac{x_n}{l^n} < \left(\frac{l+1}{2l}\right)^n \to 0 \quad \text{as } n \to \infty,$$

since $\dfrac{l+1}{2l} < 1$. Thus, $\lim\limits_{n\to\infty} \dfrac{x_n}{l^n} = 0,\ \forall\, l > 1$. □

Problem 3.16. *Suppose that* $x_n > 0$. *If* $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} = 1$, *prove* $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{\sum\limits_{k=1}^{n} x_k} = 1$.

Proof. It is clearly that $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{\sum\limits_{k=1}^{n} x_k} \ge \overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} = 1$. Hence, we only need to prove the inverse inequality. Since $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{x_n} = 1$, we know for any $l > 1$, there exists a $N \in \mathbb{N}$, such that $\forall\, n > N$, there is $\sqrt[n]{x_n} < l$. Then

$$\begin{aligned}
\sum_{k=1}^{n} x_k &\le \sum_{k=1}^{N} x_k + \sum_{k=N+1}^{n} l^k \\
&\le \sum_{k=1}^{N} x_k + \sum_{k=1}^{n} l^k \\
&< \sum_{k=1}^{N} x_k + \frac{l^{n+1}}{l-1}.
\end{aligned}$$

Hence

$$\overline{\lim_{n\to\infty}} \sqrt[n]{\sum_{k=1}^{n} x_k} \le \overline{\lim_{n\to\infty}} \sqrt[n]{\sum_{k=1}^{N} x_k + \frac{l^{n+1}}{l-1}} = l.$$

Since $l > 1$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{\sum\limits_{k=1}^{n} x_k} \le 1$. Then we are done. □

Problem 3.17. *Suppose that* $x_n > 0$, $\lim\limits_{n\to\infty} \dfrac{x_n}{n} = 0$ *and* $\overline{\lim\limits_{n\to\infty}} \dfrac{1}{n}\sum\limits_{k=1}^{n} x_k < +\infty$. *Prove that* $\overline{\lim\limits_{n\to\infty}} \dfrac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$.

Proof. Since $\lim\limits_{n\to\infty} \dfrac{x_n}{n} = 0$ and $L := \overline{\lim\limits_{n\to\infty}} \dfrac{1}{n}\sum\limits_{k=1}^{n} x_k < +\infty$, we know $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that $\forall\, n > N$, there is

$$\frac{x_n}{n} < \varepsilon \quad \text{and} \quad \frac{1}{n}\sum_{k=1}^{n} x_k < L + 1.$$

Then for $\forall\, n > N$,

$$\begin{aligned}
\frac{1}{n^2}\sum_{k=1}^{n} x_k^2 &= \frac{1}{n^2}\sum_{k=1}^{N} x_k^2 + \frac{1}{n^2}\sum_{k=N+1}^{n} x_k^2 \\
&< \frac{1}{n^2}\sum_{k=1}^{N} x_k^2 + \frac{1}{n^2}\sum_{k=N+1}^{n} x_k \cdot k\varepsilon \\
&\le \frac{1}{n^2}\sum_{k=1}^{N} x_k^2 + \left(\frac{1}{n}\sum_{k=N+1}^{n} x_k\right)\varepsilon \\
&< \frac{1}{n^2}\sum_{k=1}^{N} x_k^2 + (L+1)\varepsilon.
\end{aligned}$$

Hence

$$\overline{\lim_{n\to\infty}} \frac{1}{n^2}\sum_{k=1}^{n} x_k^2 \le \overline{\lim_{n\to\infty}} \left(\frac{1}{n^2}\sum_{k=1}^{N} x_k^2 + (L+1)\varepsilon\right) = (L+1)\varepsilon.$$

Sicne $\varepsilon > 0$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \dfrac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$, i.e. $\lim\limits_{n\to\infty} \dfrac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$. □

Problem 3.18 (3.8)**.** *Calculate the following limitations.*

(6) $\lim\limits_{t\to 1}(1-t)\tan\dfrac{\pi t}{2}$;

(8) $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{\tan x - 1}{x - \frac{\pi}{4}}$;

(9) $\lim\limits_{x\to 0} \dfrac{\cos(n\arccos x)}{x}$ (*n is odd*).

Solution. (6)

$$\lim_{t\to 1}(1-t)\tan\frac{\pi t}{2} = \lim_{t\to 1}\frac{1-t}{\cos\frac{\pi t}{2}} = \lim_{t\to 1}\frac{1-t}{\sin\frac{\pi}{2}(1-t)} = \frac{2}{\pi}.$$

(8)

$$\lim_{x\to\frac{\pi}{4}}\frac{\tan x-1}{x-\frac{\pi}{4}}=\lim_{x\to\frac{\pi}{4}}\frac{1}{\cos x}\cdot\lim_{x\to\frac{\pi}{4}}\frac{\sin x-\cos x}{x-\frac{\pi}{4}}=2\lim_{x\to\frac{\pi}{4}}\frac{\sin(x-\frac{\pi}{4})}{x-\frac{\pi}{4}}=2.$$

(9)

$$\begin{aligned}\lim_{x\to 0}\frac{\cos(n\arccos x)}{x}&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin(\frac{n\pi}{2}-n\arccos x)}{x}\\&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin\left(n(\frac{\pi}{2}-\arccos x)\right)}{x}\\&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin\left(n(\arcsin x)\right)}{x}\\&=(-1)^{\frac{n-1}{2}}n,\end{aligned}$$

where we have used $\arcsin x+\arccos x=\dfrac{\pi}{2}$. □

Problem 3.19 (3.9(4))**.** *Calculate the following limitation.*

$$\lim_{x\to\infty}\left(\cos\frac{a}{x}\right)^{x^2}\qquad(a\neq 0).$$

Solution. Note that

$$\begin{aligned}\lim_{x\to\infty}\left(\cos\frac{a}{x}\right)^{x^2}&=\lim_{x\to 0}(\cos ax)^{\frac{1}{x^2}}=\lim_{x\to 0}(1+\cos ax-1)^{\frac{1}{x^2}}\\&=\lim_{x\to 0}\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{x^2}}\\&=\lim_{x\to 0}\left(\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}\right)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}.\end{aligned}$$

Since

$$\lim_{x\to 0}\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}=e,$$

we know that for $\forall\,\varepsilon>0$, there exists a $\delta>0$, such that $\forall\,x:|x|<\delta$, there is

$$e-\varepsilon<\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}<e+\varepsilon.$$

Hence, we have for $|x|<\delta$ that

$$(e+\varepsilon)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}<\left(\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}\right)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}<(e-\varepsilon)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}.$$

Then

$$(e+\varepsilon)^{\frac{-a^2}{2}}\leq\varliminf_{x\to 0}\left(\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}\right)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}$$

$$\leq \overline{\lim_{x\to 0}} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}} \right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}}$$

$$\leq (e-\varepsilon)^{\frac{-a^2}{2}} .$$

Since ε is arbitrary, we have

$$\underline{\lim_{x\to 0}} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}} \right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = \overline{\lim_{x\to 0}} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}} \right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = e^{\frac{-a^2}{2}},$$

i.e.

$$\lim_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}} \right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = e^{\frac{-a^2}{2}}.$$

□

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